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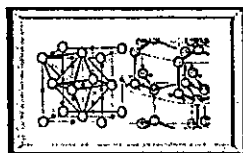
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COLD FUSION TIMES

ITALIANS, US NAVY, FRENCH (CEREM), AND OTHERS CONFIRM FLEISCHMANN AND PONS

SOME HIGH QUALITY POSITIVE RESULTS

There are important new, quality, positive reports of cold fusion. This issue of the **COLD FUSION TIMES** features reports from the US Navy (two), CEREM (The French Atomic Energy Commission), Italy, and China which supplement detailed ICCF-6 meeting summaries. The latter present material data from Japan, helium production data from Italy, and discussions of the confirmations from France, China, Japan, Illinois, and elsewhere. Some of the information has demonstrated low energy ionizing emissions and helium production linked to the excess heat. Mr. Lonchampt and others (CEREM), have continued their program to duplicate Ms. Fleischmann and Pons, and claim up to 150% excess heat. This achievement is important to note not only because of the credibility of the French Atomic Energy Commission but also because once again, Europe seems to scoop the US of own technology wrought on its own soil. More on this report (pages 9,10).

ICCF-6: REPORTS & ANALYSIS

Toya, Hokkaido, Japan The ICCF-6 conference was held at Toya, Hokkaido, Japan from October 13-18, 1996. There were 176 participants who delivered 120 papers, over-50% of which were from Japan. Scores of others were there representing 19 countries. The distribution was as follows: U.S., 37; Italy, 15; Russia, 8; France, 6; China, 4; German, 3; Korea, 2; Switzerland, 2; Austria, Canada, Hungary, India, Spain, ROC, UK, and the Ukraine, 1. The conference's six sessions many showed experimental results, some of which are presented in this issue of the **COLD FUSION TIMES**.

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HIGH POWER μ S PULSED ELECTROLYSIS USING LONG AND THIN Pd WIRES IN VERY DILUTED LiOD-D₂O SOLUTION: ANOMALOUS EXCESS HEAT

F. Celani, A. Spallone, P. Tripodi, D. Di Gioacchino
S. Pace, V. Di Stefano, A. Mancini
INFN, Lab Nazionali Di Frascati, EURESYS, ORIM

Frascati, Rome, Macerata Italy We have tested thin and long pure Palladium wires (mainly, diameter of 100 μ m, length of 160 cm), using both HV-DC electrolysis and high power-high frequency electrolysis peak current up to 25A, peak voltage up to 270V, pulse width between 200 ns and 5000 ns, repetition rate between 100 Hz and 50.000 Hz) in a 0.25 mN LiOD-D₂O solution. The experimental set-up consists of a PTFE cylinder (4 cm of diameter), a Pd wire turned around it and a Pt wire (1 mm of diameter) turned in the same way at 1 cm (constant distance) from the Pd wire. This device is located in a graduated cylindrical glass (chemical grade) filled with the electrolytic solution. A specifically designed electronic circuit has been developed to produce a very high frequency-high voltage electrolysis.

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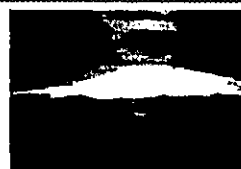
HEAT AND HELIUM MEASUREMENTS USING PALLADIUM AND Pd ALLOYS IN D₂O

Melvin H. Miles, Kendall B. Johnson, M. Ashram Imam
Naval Air Warfare Center Weapons Division, Naval Research Laboratory

China Lake, CA, Washington, DC Our best experiments produced up to 30% excess heat. 0.52 watts of excess power, and 1,400 KJoules of excess enthalpy. This amount of excess enthalpy is difficult to explain by any chemical reaction. We have demonstrated that any recombination of the D₂ and O₂ electrolysis gases in our experiments can be readily detected and easily corrected. There was never any measurable combination when the palladium cathodes were fully submerged in the D₂O-LiOD electrolyte. Our studies provide overwhelming evidence that the anomalous effects in deuterated systems are real. Nevertheless, we have not been able to solve the reproducibility problem. This research area will remain highly controversial until reproducibility can be demonstrated. The lack of reproducibility stems mainly from an unknown and uncontrolled variable contained within the palladium stock. There is a remarkable correlation of excess power with the source of the palladium.

Cont. page 4

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COLD FUSION TIMES
reports the scientific details and world-wide reports of ICCF-6



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TOWARD THE FUTURE

2

Due to information-dense coverage of the ICCF-6, the COLD FUSION TIMES's "TOWARD THE FUTURE" encompasses three important reviews of ICCF-6. The three parallax views begin below, and on the next page. These are important excerpts from the detailed writings of Jed Rothwell (further info in *Infinite Energy* or <http://fourworld.com-puserve.com/homepages/JedRothwell/>) and Elliot Kennel (confer *Secolo Scienza e Tecnologia*, Vol VII No 4).

CONFERENCE REPORT: ICCF-6 Elliot Kennel

Sapporo Japan -- I attended the 6th International Conference on Cold Fusion 12-18 October 1996. My summary opinion is that we may have seen a turning point in that the attention of the conference is being drawn towards the general realization that nuclear physics may be surprisingly different in the solid state than in gas or liquid systems. In addition, there have been several interesting innovations which suggest the possibility of improving the reproducibility of anomalous results.

1. Evidence suggesting the existence of an anomaly. There were several papers which I felt provided strong evidence for the existence of a nuclear anomaly. Among the strongest were:

a. Kasagi et al.'s study of accelerator irradiation of metal targets with low energy deuterons. In this case, the detection of up to 18 MeV protons is not at all expected, as this can not occur from normal DD fusion through the normal $P + T$ reaction pathway, which results in only 3.02 MeV protons. Kasagi indicates that this may be due to three body reactions, despite the fact that the cross section is six orders of magnitude too small for this to occur.

b. Roussetski (Lebedev Institute, Moscow), claims to have verified Lipson's observation of reproducible charged particle and neutron emission at anomalous energies. Plastic scintillation detectors, Silicon Solid State Detectors and CR-39 detectors all obtained the same results. The experiment involved a Ag-PdO-Pd-PdO-Ag heterostructure cathode which outgasses under moderately high temperature ($T < 100$ C). Of particular interest is the observation that neutron amplification occurs (200 neutrons out for one in) under certain conditions. This also suggests group behavior (coupling) of large numbers of lattice atoms combining nuclear and chemical states. The results are far beyond experimental error with thousands of excess counts.

c. I am very impressed by the claims by Miley et al., Mizuno et al., Savvatimova et al., Celani et al. and others concerning the presence of anomalous chemical species in isotopically enriched form on electrolysis cathodes. Certainly, chemical contaminants are expected on the cathode.

Cont. page 6

OVERALL IMPRESSIONS OF ICCF-6 Jed Rothwell

At this conference, the French Atomic Energy Agency described their successful replication Pons and Fleischmann's 1992 boil-off results. They got repeated, large bursts of excess heat with a performance profile remarkably similar to the original experiment. SRI also made a concerted effort to replicate this experiment, and they reported similar success.

FLEISCHMANN RETIRED, ALIENATED

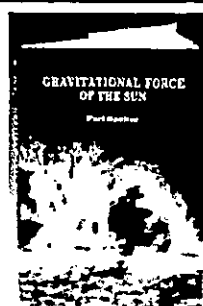
Martin Fleischmann has retired from active research and returned to England. He was originally scheduled to give a one-hour keynote lecture at the conference, but this was canceled. I asked him why. He replied he "has already said everything there is to say." He feels alienated from mainstream research in the field, which he thinks is politicized. He feels that people have not listened to his advice and ideas.

OKAY, SOME GOOD NEWS

In my opinion the two best papers were from Miley and Pons. G. Miley, "Experimental Observations of Massive Transmutations Occurring in Multilayer Thin-Film Microspheres After Electrolysis" - This was similar to Miley's lecture and paper given at the Second International Low Energy Nuclear Reactions Conference (ILENR2). Miley showed some additional data strengthening the observations of three zones of transmutation. He explained that zones are characteristic of fission reactions. He showed data for beads with glass cores. These produced little heat and only a small number of apparent transmutations, with large sections of the spectrum flat compared to the plastic core beads.

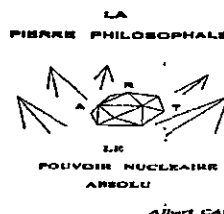
In conversation, he explained that the make up of the beads was different in each of the twenty runs. Some came from CETI while others were fabricated at the University of Illinois. Some had multiple layers of nickel and palladium, some were nickel only, and thickness was varied. He said he sees no point in doing the same experiment over and over. He wants to explore a variety of materials and thin film configurations. Several companion papers about the CETI device were scheduled for the same hour that Miley talked, but none were delivered.

Cont. page 7



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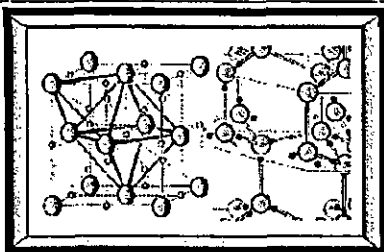
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Page 4 - Dieter Britz (his web page URL = <http://kemi.aau.dk/~britz/>)
Back Cover: Europa - surface highly disrupted by fractures and ridges [238 x 225 kilometers]. Symmetric ridges in the dark bands may be spreading centers in ocean basins. - Jet Propulsion Laboratory
Cold Fusion Times LOGO - Palladium must be fully filled with deuterons to generate the reactions. In the logo, the face-centered cubic fully loaded metallic structure is shown on the left, totally filled with deuterons obtained from the heavy water on the right hand side.



COLD FUSION TIMES

*The Journal Of The Scientific Aspects Of
Loading Isotopic Fuels Into Materials*

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EXCESS HEAT IN (D/Pd) GAS-LOADING

Xing Li, Wei Yue, Gui Huang, Hang Shi, L. Gao, M. L. Liu
Dept. Physics, Tsinghua University

Beijing, China - The "heat after death" has revealed a fact that the electrolysis is not necessary for "excess heat" phenomena. Provided that enough deuterons are filled in the palladium crystal lattice, we may expect to see the "excess heat" in a gas-loading system also. The Italian, Indian, and American scientists have succeeded in loading the deuterium gas into the palladium sample with loading ratio (D/Pd) greater than 0.80 under the pressure less than 1 atm. This facilitates the quantitative observation of the "excess heat" in a gas-loading system. Besides, the gas-loading system has showed high reproducibility in loading, and has the advantage to be operated at high temperature which means a higher efficiency as an energy device hence, a series of gas-loading experiments have been conducted to observe the "excess heat" in a gas-loading system.

The observation of the "excess heat" would be presented in a system with the following 4 features:

(1) Heat treatment of the palladium surface in a closed system which allows the annealing in vacuum or in deuterium (hydrogen) gas with the capability of controlling the temperature precisely. Particularly, the palladium surface would not be

exposed to the air (oxygen) before the "excess heat" measurement.

(2) Monitoring the loading ratio in situ with both the volumetric method and the electrical resistance method. Then the correlation between the "excess heat" and the loading ratio would be monitored.

(3) Measuring the "excess heat" with an accuracy of 0.2 W in a long period (days to a month). Since the volume of the palladium wire is about 0.2 c.c., we are supposed to see the "excess heat" if its power level is above 1 Watt per c.c. palladium.

(4) Accumulating the nuclear products (if any) in a closed stainless steel vessel. Helium-3, helium-4, and tritium are supposed to be checked first after the long run using mass-spectrometric method and nuclear method

A pair of "twin system" have been run in parallel with deuterium gas and hydrogen gas using "twin palladium samples". "Twin samples" means that they were cut from the same roll of the palladium wire cleaned, heated and cooled under the same condition, in fact they were born as a pair of "twin". The preliminary runs have shown distinct contrast between these "twin samples", which may be explained as an evidence of "excess heat" in D/Pd system. A series of runs are expected before the ICCF-6.

ENGINEERING REPORT FROM ICCF-6

I. Steps backward

1. Kunimatsu reported a negative result on excess heat using a flow calorimeter version of his fuel cell experiment. This is important as his report at the last conference was positive. These experiments were done at high loading, more than 0.95 according to Kunimatsu (according to Michael McKubre, their resistance calibration is different that what SRI uses, so that these results may be equivalent to SRI "border region" conditions where things don't work very often).

2. Srinivasan no longer believes his light water excess heat results. SRI did not find excess heat in his experiments, and the group at Bose Corp explained his results as recombination. Operationally, he is retracting his Ni light water excess heat results.

II. Results neither good nor bad

1. The New Hydrogen Energy lab (NHE), reported positive results (up to 18 %

excess heat) in a fuel cell experiment. When they tried to repeat it in a flow calorimeter, they got no excess. They noted that the mass flow calorimeter gives quite a different temperature history, which they feel is important. I go the impression that they wanted to re-do their mass flow calorimeter to have a variable temperature capability.

2. KEK built up a helium detection system that can measure He-4 at 17 parts per trillion. Very impressive, but they have not gotten a recent positive result to test it yet.

III. New results

1. Preparata reported getting a factor of 2 excess heat using a radical type of electrolysis system. Fleischmann previously suggested using electromigration to drive the deuterium in a cathode so as to increase the loading. The U. of Milan group puts a few hundred volts across a 50 micron diameter and few cm long cathode, and runs electrochemistry at a few hundred volts in 10^{-5} molar LiOD.

2. Gozzi has observed 90 keV gamma spots on film from a heat-producing cathode. This was a single recent observation, but appeared to be pretty solid.

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OF GREAT INTEREST
TO OUR READERS
SEE FOR THE WORLD (PAGE 8)

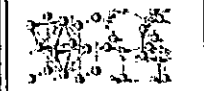
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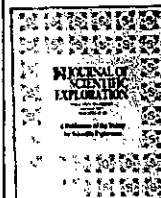
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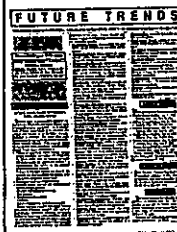
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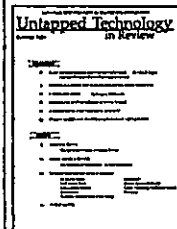
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continued on page 6



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HEAT AND HELIUM MEASUREMENTS USING PALLADIUM

M. Miles, K. Johnson, M.A. Imam Continued From Page 1

The best reproducibility was obtained using Pd-B materials supplied by the Naval Research Laboratory. Seven out of eight experiments that used Pd-B cathodes produced excess power. A high success ratio was also obtained using Johnson-Matthey materials. Seventeen out of twenty-seven experiments that used this palladium source produced excess heat, in contrast there were several palladium sources that never produced excess power in any experiment. Our calorimetric results and problems are practically identical to those reported by the SRI laboratory. They are also consistent with many other laboratories that have reported excess heat. Calorimeters that are capable of detecting excess power levels of 1 watt per cubic centimeter of vanadium are essential for research in this field. The small volume of palladium in co-deposition experiments is likely the reason why it was difficult to detect excess power effects in such studies.

Our results indicate that helium-4 is the missing nuclear product. Thirty experiments have shown a correlation between either excess power and helium production or no excess power and no excess helium. Studies using both glass and metal flasks place the 4He production rate at 10^{11} to 10^{12} atoms/second per watt of excess power. This is the correct magnitude for typical fusion reactions that yield helium as a product. It is highly unlikely that our heat and helium correlation could be due to random errors. The only valid experiments that showed significant excess power but no excess helium involved a Pd-Ce cathode. The odds are less than one in a million that our complete set of thirty-three heat and helium results could be obtained from random experimenters errors. ➤

HIGH POWER μS PULSED ELECTROLYSIS USING LONG AND THIN PD WIRES IN VERY DILUTED $\text{LiOD-D}_2\text{O}$ SOLUTION: OBSERVATION OF ANOMALOUS EXCESS HEAT

F. Celani, A. Spallone, P. Tripodi, D. Di Gioacchino, S. Pace, V. Di Stefano, A. Mancini continued from page 1

Read-out circuits are linked to a PC to acquire cyclically (every 8 seconds) the signals from the sensors. Regarding the heat measurements, we used a flow calorimeter. A Joule heater, switched on and off cyclically every 3 hours, has been overimposed, in order to continuously calibrate the apparatus. In some previous exploration experiments (with a simple bath-calorimeter), probably because of the particular cathode geometry specifically developed in order to enhance the Electromigration Effect), the excess heat reached the value of about 70 W for several hours (about 200% of the input power). Moreover, correlation has been found between the excess heat and the electrical resistivity (by AC measurements) of the Pd-D wire. Systematic studies, with the flow calorimeter previously described, are in progress in order to reproduce the results using a new wire. ➤

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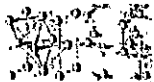
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Journal Articles Which You May Have Missed

Some comments from Dieter Britz -
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Good II WR; *Fusion Technol.* 30 calculated, for (1996) 132. "Comments on electrolysis at 'Calorimetry, excess heat, and Faraday efficiency in Ni-H₂O electrolytic cells". W.R. Good, of the Hydrocatalysis Power Corp., as electrolysis efficiency less than 100%.

from the Bose lab, by Shkedi et al. Good reckons that the Bose team did not follow recommended procedure and for this reason failed to detect hydrinos as proposed by Mills.

Savvatimova IB, Karabut AB; *Poverkhnost'* 1996 (1) 76 (in Russian), "Radioactivity of palladium in glow discharge". Foils of Pd, Nb, As and Ti and other metals were subjected to a glow discharge (10-50 mA/cm², 100-500V) in hydrogen, deuterium or a mixture of the two, and afterwards the radioactivity of the foils was measured as a function of time. When using deuterium, there was radioactivity, decaying with a half life of 13.8 h, which the authors conclude to be consistent with the decay of ¹⁰⁹Pd, light water shows only the background by fusion. The level was 5-10 times that of the background.

Shkedi Z, *Fusion Technol.* 30 (1996) 133. "Response to 'Comments on 'Calorimetry, excess heat, and Faraday efficiency in Ni-H₂O electrolytic cells'". In response to a polemic by Good (1996) of Hydrocatalysis Power Corp, charging the Bose team (Shkedi et al) with not following recommended procedures, Shkedi responds that they did indeed follow them, and that they did find apparent excess heat initially. When they however examined the results they found that the shortfall from 100% of the Faradaic efficiency of their electrolysis was able to fully eliminate the excess heat. He further comments on the reason for the high excess heat percentage claimed by Good; this being caused by the way it is

Stukan PA, Rumyantsev YuM, Shishkov AV, *High Energy Chem.* 27 (1993) 461. (Orig. Russian in *Khim. Vys. Energii* 27(6) (1993) 65). "Generation of hard radiation and accumulation of tritium during electrolysis of heavy water".

In 1990, this team began the experiment described here. They electrolyzed a 2% solution of Li₂CO₃ in heavy water at cathodes of Pd and Ti, both 4 mm dia. rods and measured hard radiation given off over time, using a beta-type scintillator and photomultipliers. The cell current was 1A/cm². This showed a radiation sequence with time, roughly 10-20 times in counts/s of the sequence before the current was turned on. A control run with light water shows only the background itself. By using a paraffin shield and noting the effect, they were able to state that the radiation consisted largely of neutrons, and estimated the flux to be about 2×10^3 n/s. They also measured tritium accumulation in the cell, by removing aliquots repeatedly and, after an initial quiescent period, there was a steady, roughly linear rise in the amount of tritium produced vs time, somewhat greater for the Pd cathode than for the Ti one. With the current off, or current on with light water, much smaller amounts of tritium were seen; in the case of light water, the team suspects tritium coming out of the Pd from previous heavy water runs. The tritium production on Pd in heavy water translates into about 2×10^4 t/s, 5 orders of magnitude larger than the neutron flux. They note the discrepancy but do not attempt an explanation. No unexpected heating of the cell was observed. ➤

ENGINEERING REPORT FROM ICCF-6

continued from page 3

3. Gozzi appears to have a reasonably solid time-correlation between excess heat and He-4 production. Atmospheric contamination is monitored through Ne mass spectroscopy, and the new results appear to be free of such contamination. The signal to noise is good, and the He-4 is somewhat short of being quantitative (it goes between 5% and 50% assuming 24 MeV per reaction -I think that a lot

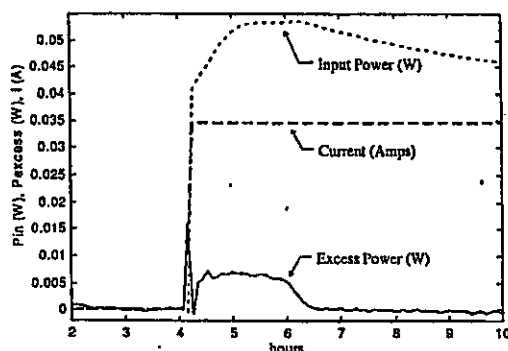
Anomalous Effects in Deuterated Systems

Melvin Miles, Benjamin Bush,
Kendall Johnson
Naval Air Warfare Center Weapons Div.

China Lake, CA Excess power was measured in 28 out of 94 electrochemical experiments conducted using palladium or palladium-alloy cathodes in heavy water. Reproducibility continues to be the major problem in this controversial research area. Based on our experiments, this lack of reproducibility stems from unknown variables in the palladium metal. The best reproducibility for excess power was obtained using palladium-boron materials supplied by the Naval Research Laboratory. Our basic isoperibolic calorimeters were capable of measuring excess power with a sensitivity of $\pm 1\%$ of the input power or ± 20 mW, whichever was larger. Calorimeters that are capable

of detecting excess power levels of 1 watt per cubic centimeter of palladium are essential for research in this field.

Results from our laboratory indicate that helium-4 is the missing nuclear product accompanying the excess heat. Thirty out of 33 experiments showed a correlation between either excess power and helium production or no excess power and no excess helium. The collection of the electrolysis gases in both glass and metal flasks place the helium-4 production rate at 10^{11} to 10^{12} atoms per second per watt of excess power. This is the correct magnitude for typical deuterium fusion reactions that yield helium-4 as a product. Anomalous radiation was detected in some experiments by the use of X-ray films, Geiger-Mueller counters, and by the use of sodium iodide detectors. There was never any significant production of tritium in any of our experiments.



Measurement of the Exothermic Heat of Absorption of Deuterium Into a Palladium wire (1mm x 4.3 cm) Using the New Calorimeter Developed at China Lake Power (Watts) as a function of time [after Miles, Bush, Johnson, NAWCWPNS TP8302]

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Lavoisier melting a "worthless" platinum nugget

of the helium stays in the cathode, so they don't see it in the flow stream.)

4. Okamoto considered the importance of deuterium flux in excess heat experiments, as found by SRI last year. He optimized loading using a series of loading/deloading current cycles, and then looked for heat and neutrons. He appears to see a persistent and reproducible excess heat effect that is small (2-3%) but well within the accuracy of his system. He appears also to see a small excess of low energy neutrons that are correlated with the heat.

5. An effort associated with the French Atomic Energy Commission was reported by Longchamp. They attempted a replication of the Pons-Fleischmann boil-off experiment. In the majority of their experiments, no excess heat was observed. However, in some cases, they obtained a significant 2-3 per cent effect prior to boil off, and a factor of about 2 excess during boil off. No "heat after death" was observed. This is the first confirmation of this effect.

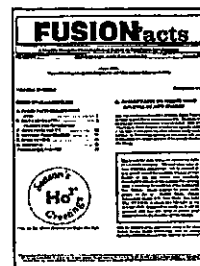
6. An important experiment was carried reported by a collaboration between Roussetski at the Lebedev Institute and Lipson from the Institute for Physical Chemistry. A 30 micron thick Pd cathode was exposed to an oxygen flame (1000 C for I think 30 seconds) to make PdO on both surfaces; one surface was coated with gold. The foil was loaded electrochemically by Lipson, and then brought to the Lebedev, where it was heated to 40 C in front of a variety of detectors. Various emissions were observed: protons and alpha particles were seen in the MeV range, detected using Cr-39, plastic scintillation, and also with a silicon surface barrier detector.

7. Mike McKubre reported on difficulties in getting high loading and keeping it at high current densities. He pointed out that in his earlier work with positive results, that the cathodes would load to more than 0.90, and then hold their loading up to 500 mA/cm². During the last two years, the cathodes typically begin de-loading near 100-200 mA/cm², so that there is no high loading at high current levels. Solving this problem is a prerequisite for getting heat in his view. However, Mike showed an excess "cold" burst, where there appears to be an anomalous cooling event on the order of 5 KJ that is significant. This is the first such report.

cont page 6

FUSION FACTS FOLDS

Salt Lake City, Utah - After seven years, the first and oldest of the cold fusion journals has folded. With the December 1996 issue, *Fusion Facts*, edited by Hal Fox with the help of Robyn Harris and Dineh Torres, will be its last. Their efforts will focus on the quarterly *Journal of New Energy*. The staff of the COLD FUSION TIMES wishes them well as *Journal of New Energy* enters its second year. We all thank Hal, Robyn and Dineh for their effort at informing both this community and those less knowledgeable about cf.



The final cover of *Fusion Facts*

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COLD FUSION TECHNOLOGIES

ENGINEERING REPORT FROM ICCF-6

Continued from Page 5

There were several other results of note -Claytor and colleagues have recently improved their results with a Pd/Rh/Co alloy; Romodanov has seen a huge excess tritium effect (a big increase in his already huge tritium effect) by adjusting the relative H/D ratio. Takahashi's group has some nice results correlating heat with products. Notoya thinks she sees gammas from her light water excess heat experiments.

Miley and others talked about massive transmutations. In the CETI cells, the excess heat (if real) would correspond to a significant fraction of product isotopes. So Miley has used SIMS, EDX, and neutron activation to search for products. He sees a vast array of elements, with on the order of 40% of the Ni perhaps being affected. The

spectrum of proposed "products" fall into four major peaks as a function of Z, with many elements heavier than Ni appearing in Ni beads. His results appear to be very similar to those of Mizuno.

Savvatimova has claimed a similar effect for years, and Collis reports that a colleague of Piantelli in Italy is getting a similar effect. If these results are correct, they point to a process in which heavy nuclei are somehow reacting. Many in the community think that the results are due to contamination, but George believes that there are no sources for such contamination in his cell. In addition, the NAA results suggest modest (but significant) isotopic anomalies away from natural abundances in Cu and Ag isotopes (4-8% deviations from natural abundance).

>

CONFERENCE REPORT:

ICCF-6

Elliot Kennel
cont. from page 2

However, isotopically enriched crud is a different story. To a lesser extent, it is impressive that the observers report very similar four-humped elemental distributions, reminiscent of the two-humped fission fragment distribution curves. This is not expected for random dirt.

d. Ed Storm's measurements of open circuit voltage as a function of loading suggest that a previously unknown gamma phase in the Pd-D system appears at loadings above 0.85, which is associated with anomalous excess heat measurements.

2. **Innovations.** Electromigration appears to be a significantly superior method to produce excess thermal energy, as Celani et al report reproducible excess heat results with relatively large output (>100% in

many cases). The basic idea is to use a pulsed electric field to drive deuterium toward the center of a wire once it has been loaded electrochemically. It is very reasonable to expect high local loading using this technique, and perhaps it is really responsible for enhancing anomalous results.

Another Italian group (DeNinno et al) has produced a technique for achieving high loading of deuterium while avoiding a phase transition. This requires high pressure, high temperature deuterium. Once loaded with this technique, subsequent loading electrolytically is much easier.

This method may significantly enhance the performance of existing cathodes which have not loaded sufficiently well at low temperature (i.e., in a flow calorimeter). In addition, loading occurs much more rapidly after the pretreatment. >

GUEST BOOK REVIEW

This issue by Hal Fox

"The Coming Energy Revolution"

Jeane Manning

[Avery Publishing Group, Garden City Park, New York, 1996, 230 pages, ISBN 0-89529-713-2]

This timely book is highly recommended for all readers, especially for the general reader. It provides both an excellent report on the historical background of the search for new energy and introduces the readers to those who are now involved in various aspects of new energy developments. The book is divided into four parts: Part I - Past Revolutionaries; Part II - Space Energy and the New Physics; Part III - Emerging Energy Technologies; and Part IV - The Energy Revolution -- Potential Amid the Problems. In addition, the book begins with a favorable Foreword by Brian O'Leary, who is well known in the new-energy field.

Part I discusses some of the historic work of Nikola Tesla, John Ernst Worrell Keely, Walter Russell, Thomas Henry Moray, Lester Hendershot, Viktor Schauberger, and Wilhelm Reich.

Part II tells us about some of today's geniuses such as Moray King, Harold Puthoff, and Thomas Bearden and their efforts to help provide a scientific underpinning for "space energy." Next, the works of Kenneth Shoulders and his high-density charge clusters, Wingate Lambertson and his cermet devices, and John Hutchison and his new rock-crystal devices. These three innovators are apparently tapping space energy with three different solid-state devices. The work of the late Floyd "Sparky" Sweet is described. Four other persons who have worked with rotating machines in the production of energy are introduced: Bruce dePalma (now of New

Zealand), Bertil Werjefelt (Hawaii), Shuji Inomata (Japan), and Paramahansa Tewari (India). These are four of the leading lights in the development of the "N-Machine" which was instigated by Bruce dePalma.

Part III discusses the advent of cold nuclear fusion, first announced in 1989 by Professors Stanley Pons and Martin Fleischmann at a press conference called by the staff of the University of Utah. The author cites that the work not only is continuing but is being commercialized. The work of the late Francisco Pacheco in producing hydrogen gas from sea water and using this power for a variety of energy devices is presented. In a similar task to develop the power of hydrogen, Jeane Manning brings us up to date with the life's work of Roger Billings (formerly from Utah and now in Independence, Missouri).

Part III also includes the stories behind the work of Peter Lindemann and George Wiseman in using relatively low-grade heat as a useful power source. Their efforts to develop Low-Temperature Phase Change (LTPC) technologies are discussed. Also, Jeane tells us the story of Harold Aspden (one of our active correspondents) and his search for better ways of using magnets and heat to produce usable energy. This part of the book includes a report on William Baumgartner who is improving on the work of the late Viktor Schauberger in the use of vortexing water.

The trials and tribulations of the use of the new technology of low-flow water power, as developed by Martin Burger, is included in this part of Manning's book. Finally, the work of Johann Grander (another follower of Viktor Schauberger's work) and his discovery of "living water," the rotating energy-producing disks of Paul Baumann at the Methernitha commune, and the hydrosonic pump of James Griggs, are discussed.

These are fascinating tales of the difficulties of getting (or not allowing) new-energy developments to get to the market place. >

Authors, scientists, writers, inventors, entrepreneurs, consultants, economists, futurists - you're invited to submit manuscripts for consideration for publication in the

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COLD FUSION TIMES

ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

Σ SCIENTIFIC PAPERS AT ICCF-6

a. Lonchampt et al. [The French Atomic Energy Commission] duplicate P&F, 150% excess heat.

b. Dufour et al. [Laboratoire des Science Nucleaires Conservatoire National des Arts et Metier, France] - 8 keV per hydrogen, 16 keV per deuterium.

c. Zhang Qingfu et al. [Institute of Atomic and Molecular Science, Chengdu] - x-ray diffraction crystal structure of the Ti-D system changes from hexagonal to face-centered cubic TiD₂ when excess heat is observed.

d. Oyama et al [Tokyo University of Agriculture and Technology] - sulfur additives prolong heat after death.

e. Hagans et al [Naval Research Labs] - impurity studies using SPS, GDMS and ICP.

f. Ochiai et al [Osaka U] - anomalous high energy protons.

g. Wang et al [Institute of Modern Physics, Lanzhou China] 1-18 keV hydrogen bombardment on Pd and Ti, - anomalous x-rays, neutrons of 8×10^4 n/sec, high energy protons.

h. Itoh et al [Mitsubishi Heavy Industries] - nuclear detection expts.

i. Chen et al [Institute of Nuclear Physics, Beijing] - x-ray emission during gas discharge loading at ~26 keV.

j. Savvatimova et al. [Luch] - transmutation in Pd cathodes using glow discharge loading.

k. Zhang et al. [Institute of Applied Physics, Beijing] - x-ray generation.

l. Yamada et al [Iwate University] - occasional large neutron bursts 2700 counts in 5 seconds in a gas loaded discharge system

m. Miley et al. [U. of Illinois] - transmutations from thin film deuterium and hydrogen

n. Claytor et al. [Los Alamos] - Miley-type experiment.

o. Kozima et al [Hizuoka University] - transmutation results similar to Miley.

OVERALL IMPRESSIONS OF ICCF-6

Jed Rothwell cont. from page 2

S. Pons, "The ICARUS 9 Calorimeter: Summary of Three Years Designing, Testing and Operation of this Device at the IMRA Europe Science Center" It describes a cell that was held at boiling for three months, producing hundreds of megajoules from cathodes weighing a fraction of a gram.

P. L. Cignini, D. Gozzi et al., "X-Ray, Heat Excess and 4He in the Electrochemical Confinement of Deuterium in Palladium" Excess heat, helium and x-rays were observed. Helium was measured by drawing off samples of gas, filtering out everything but helium, and putting the gas into a QMS (quadrupole mass spectroscopy) chamber integrated into the experimental apparatus. In a four-cell experiment lasting 950 hours, more than 1000 samples were taken from each cell.

At first glance, the helium did not appear to be correlated with heat, but a sophisticated statistical analysis, taking into account the sampling periods and percent of total gas in one sample, shows that did actually appear together. The curves for helium change completely after the complex statistical massaging. X-Rays were measured with x-ray film positioned 5 cm from the cell. X-ray film placed next to a Pt blank cell showed no fogging. Film next to Pd cells that produced heat were fogged, with a shadow where the anode blocked the cathode. A sophisticated microscopic analysis of the film was performed, based on the physics of film and

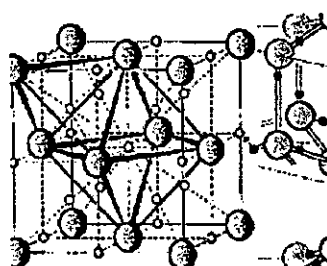
comparisons to film of the same stock exposed to known levels of x-rays. It was concluded that the "energy of radiation" (the power of the x-rays) was 89 $\pm$ 1 keV and the total energy intercepted by the film was 12.0

± 0.4 kJ. Gozzi's cathode is a bundle of palladium wires, rather than a single block of palladium. This increases surface area.

E. Botta, "Further Measurements on 4He Production from PdD₂ Systems in Gas Phase" The Italian university and INFN (Italian government Nuclear Physics Laboratories) showed more evidence for helium production in one run. They use an interesting technique to achieve high loading. Deuterium gas is loaded palladium plate under pressure, and "a constant electric field of a few hundred millivolt per cm" is applied from one edge of the plate to the other, which moves the deuterons across it by electromigration. The ends of the plate are sealed with gold, and I think this causes the deuterium to "pile up" against the ends. This electromigration technique was used by several other Italian workers to enhance the CF effect, and it is the basis of the proton conductors used by Mizuno, Biberian (ICCF5), Oriani and Karabut (P-001).

With a proton conductor, you charge both sides of the plate (rather than the ends). If you leave a direct current charge running continuously, one side of the plate would become highly loaded with deuterons but the other side would not, so you use a square wave current that alternates every minute or so, charging first one side, then the other. Mizuno and others speculate that this also causes the deuterons to "slosh around" inside the conductor, occasionally piling up in local spots at a loading ratio much higher than the average for the entire conductor, which causes a powerful CF reaction in that isolated spot.

コルト フュージョン タイムズ



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*The Journal Of The
Scientific Aspects Of Loading
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Continued on page 8

**Σ SCIENTIFIC PAPERS AT
ICCF-6 cont. from page 7**

p. Mizuno et al. - transmutation and anomalous isotope production.

q. Ohmori et al. -transmutations

r. Notoya et al [Hokkaido University] - transmutations.

s. Vysotskii et al. - controlling the nuclear decay constants of radionuclides by restricting the modes of Mossbauer nuclei.

t. Vysotskii et al.- production of isotopically pure Fe⁵⁷, purportedly does not exist in nature.

u. Bertalot et al [EURATOM-ENEA] report excess heat and helium.

v. Claytor et al. [Los Alamos] - excess tritium from glow discharge experiments.

w. Yuki et al. [Tohoku University] - enhanced reaction rates from low energy deuteron bombardment of metals. ➤

FOR THE WORLD

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8

(More Information -- See Pages 2 And 3 Too)

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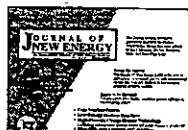


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QUALITY POSITIVE RESULTS

Continued From Page 1

Cold Fusion works. There is now overwhelming data to support this field. And real scientists know what the ingredients are: an active sample, full loading, and the ability to activate the loaded sample. This includes reproduction of the excess heats, better definition of the nuclear ash formed, and much groundbreaking regarding how -- and when -- these processes occur.

Although, the Italians, Americans, French, and Russians have relentlessly pursued this field and never gave up, Japan is reportedly on the verge of throwing in the proverbial towel, as some of the laboratories are unable to resolve the role of isothermal control in nonequilibrium systems. This is a watershed opportunity for United States technology. Hopefully, now that these reports are out, mistakes of the past won't be repeated.

Why can't some of those myopic bureaucrats open his or her eyes? When will the U.S. patent office become responsible and responsive to U.S. citizens, including inventors? When will either political party in the U.S. do their jobs? Despite it all, maybe 1997 will be the year of Cold Fusion. ➤

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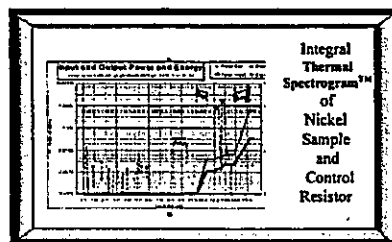
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OP-ED COLUMN --

IS JAPAN UNABLE TO MAINTAIN A TECHNOLOGIC EDGE?

These are excerpts from a longer letter written by Jed Rothwell. He is concerned that, despite the confirmations by Italy, US, France, Russia, etc., that NHE (Japan) has done more than 50 experiments in a row with no success, and IMRA (Japan) has also done 32 experiments without success. Jed maintains that they have ignored the literature and improved techniques. Although some Japanese may look with curiosity, in many ways Jed Rothwell is right, and we support his enthusiasm for this field, and the importance of doing things correctly. His full text is available at his web site (URL page 2) or in *Infinite Energy*. ➤

**Critique Of NHE Experiments
Open Letter to The NHE**

Jed Rothwell

Over the last three years I have felt a growing sense of unease about the NHE cold fusion program. At ICCF6, during the visit to the NHE lab, and in conversations with Pons, Fleischmann, Kennel, Asami and others, I found my worst fears have been justified. The program has great promise, but in execution it is deeply flawed. ..Your researchers are ignoring techniques described in papers by Ikegami, Storms, Fleischmann, Celani and other leading researchers. In the Japanese Journal of Applied Physics, Ikegami [1] spelled out four essential conditions to achieving the cold fusion effect: high current density, high loading, a cathode temperature over 80 deg C, and proper surface preparation. Celani adds a fifth essential: non-equilibrium triggering conditions. (Storms disagrees, but most other experts say a trigger is needed.) Storms [2] and Cravens [3] describe essential characteristics of the palladium, and methods of testing it. I believe you have met the first two conditions listed by Ikegami and ignored all the others. As McKubre pointed out in his closing remarks, an SRI-style flow calorimeter is designed to ensure maximum equilibrium, unchanging temperatures, and a cool cathode. At ICCF3, ICCF4 and ICCF5 Fleischmann [4, 5, 6] described the essential role of high temperatures in promoting the reaction, in what he calls "positive feedback" between the temperature and the rate of excess enthalpy generation." At ICCF4, Storms [7] characterized the role of temperature, and McKubre [8] described one of the most dramatic heat bursts in the SRI flow calorimeter, which occurred in 1993 when the flow was accidentally blocked and the cathode temperature rose higher than planned. McKubre pointed out that this confirmed Fleischmann's hypothesis.

The NHE lab is considered the flagship of cold fusion. It has the best equipment and some of the finest, most dedicated and skillful workers. You have the opportunity to make scientific history. I urge you to open your eyes to some of the

well-established, replicated alternatives before it is too late. You should start with a 20-minute taxi trip over to the University of Hokkaido. The experiments being performed there by Mizuno, Ohmori, Enyo et al. are the best in Japan, and among the best in the world. They get consistent results at high signal to noise ratios. They observe dramatic, massive transmutations that produce unnatural isotopic ratios and other indisputable proof of nuclear reactions. This proof is more convincing than the mainstream neutrons, x-rays or tritium results, because the effects are large and easy to detect, non-transient, and cumulative. The longer the reaction runs, the more metal is transmuted. Mizuno's proton conductors [9] have confirmed by Oriani, [10] who is one of the best electrochemists in the world. Enyo and Mizuno's palladium transmutation results have been confirmed by Minevski & Bockris [11]. Ohmori's [12] gold results have been confirmed by Swartz [13]. Ohmori has repeated his experiment more than 50 times, demonstrating massive excess heat and transmutations in every run. He inputs 0.1 watts. After a 3 to 5-day incubation period, he observes between 0.4 and 1.0 watts output, producing a 4 deg C Delta T temperature in his isoperibolic calorimeter. The reaction generates transmuted material at levels 100 times higher than contamination from all sources in the cell, including material with highly unnatural isotopic distribution.

You have seen dramatic examples of what can go wrong in your own lab with the Pons and Fleischmann boil-off experiment. Even with some direct, hands-on help from Stan Pons, you were not able to replicate this in 20 attempts. Both SRI [15] and the French Atomic Energy Commission (CEREM) [16] were able to replicate this experiment. Why did they succeed where you failed? I believe it is because: 1. McKubre is an experienced electrochemist; 2. Lonchampt et al. got more direct, hands-on assistance from Pons and Fleischmann over a longer period of time than you did; 3. Lonchampt insisted on doing a rigorous precise, engineering-style replication, down to the last wire. He is an engineer. He works the way an engineer would set up a semiconductor production line: he does everything exactly according to the book, making no attempt to be "creative" or to change anything. He knows that you must master the subject first, then add your own contribution to the field.

I saw several gross differences between your boil-off cell and Pons and Fleischmann's, starting with the fact that theirs was half-silvered. I have no idea which differences in the hardware or protocol caused your cell to fail, but since the experiment has now been independently replicated twice, it must be your mistake, not Pons and Fleischmann's. If you had gotten more help, and if you had replicated the experiment with exactly the same glassware and wires, or better yet, if you had practiced with a cell and cathode that had worked repeatedly in France, then your boil-off experiment would have worked as well as CEREM's did. You could have built on that lesson. You might go back and do so now, but it would be better to start with an easier, more reliable experiment like Ohmori's.

**REPRODUCTION OF
FLEISCHMANN AND PONS**

Lonchampt, Bonnetain, Hicter
continued from page 10

In the intermediate regime, the water vapor carried away by the gases of the electrolysis are large, and cannot be evaluated precisely. This makes the analysis difficult. At boiling, three positive experiments have been successful giving excess enthalpies rates of 80 % to 150 %. We present the results obtained with different types of cathode materials: pure, annealed and cold worked palladium and alloys (Pd-Rh and Pd-Ce). In conclusion, we confirm the results published by Fleischmann and Pons (1), more particularly in the boiling regime.

1. M. Fleischmann, S. Pons, Phys. Lett. A 176, 118 (1993) ➤

1. H. Ikegami, "The Next Steps in Cold Fusion," Oyou Butsuri, Vol 62, No. 7, July 1993, p. 717
2. E. Storms, "How to Produce the Pons-Fleischmann Effect," Fusion Technology, March 1996.
3. D. Cravens, "Factors Affecting the Success Rate of Heat Generation in CF Cells," Proc. 4th Int. Conf. on Cold Fusion, Lahaina, Maui, EPRI TR-104188, Vol. 2, 18-1 Voted Best Paper in Conference by Fleischmann
4. M. Fleischmann, "Calorimetry of the Pd-D₂O System: from Simplicity via Complications to Simplicity," Frontiers of Cold Fusion; Proc. 3rd Int. Conf. Cold Fusion, Nagoya, 1992, ed. H. Ikegami (Universal Academy Press, Tokyo, 1993), p. 47
5. M. Fleischmann, "Calorimetry in the Pd-D₂O System: The Search for Simplicity and Accuracy," Proc. 4th Int. Conf. on Cold Fusion, Lahaina, Maui, EPRI TR-104188, Vol. 1, p. 1-13
6. M. Fleischmann, "More about Positive Feedback; more about Boiling," Proc. 5th Intern. Conf. on Cold Fusion, Valbonne, France, IMRA Europe, p. 140
7. E. Storms, "Some Characteristics of Heat Production Using The 'Cold Fusion' Effect," Proc. 4th Int. Conf. on Cold Fusion, Lahaina, Maui, EPRI TR-104188, Vol. 2, p. 4-1
8. M. McKubre, "Loading, Calorimetric and Nuclear Investigation of the D/D₂ System," Proc. 4th Int. Conf. on Cold Fusion, Lahaina, Maui, EPRI TR-104188, Vol. 1, p. 5-52
9. T. Mizuno, "Formation of 197Pt Radioisotopes in Solid State Electrolyte Treated by High Temperature Electrolysis in D₂ Gas," Infinite Energy, #4
10. R. Oriani, "A Confirmation of Anomalous Thermal Power Generation From A Proton-Conducting Oxide," Proc. 6th Int. Conf. on Cold Fusion, Hokkaido, Japan, O-036. To be published in Fusion Technology, 1996.
11. Z. Minevski, "Two Zones of 'Impurities' Observed After Prolonged Electrolysis of Deuterium on Palladium," Infinite Energy, combined issues 5 and 6, p. 67.
12. T. Ohmori, "Isotopic Distributions of Heavy Metal Elements Produced During the Light Water Electrolysis on Au Electrode," Proc. 2nd Int. Low Energy Nuclear Reactions Conf., College Station, Texas, and Proc. 6th Int. Conf. on Cold Fusion, Hokkaido, Japan, TS-004
13. M. Swartz, "Possible Deuterium Production and Light Water Excess Enthalpy Experiments Using Nickel Cathodes," Proc. 2nd Int. Low Energy Nuclear Reactions Conf., College Station, Texas
14. G. Miley, "Nuclear Reaction in Palladium-Hydrogen System," Infinite Energy, #8
15. S. Crouch-Baker, "Mass Flow Calorimetric Studies under Non-Steady State Conditions," Proc. 6th Int. Conf. on Cold Fusion, Hokkaido, Japan, P-004
16. G. Lonchampt, "Reproduction of Fleischmann and Pons Experiments," Proc. 6th Int. Conf. on Cold Fusion, Hokkaido, Japan, O-044

**WORLD WIDE WEB SOURCES
ON COLD FUSION**

• **COLD FUSION TIMES** home page:
<http://world.std.com/~mica/cft.html>

• Dieter Britz's home page <http://kemi.aau.dk/~britz/>
John Logajan's home page:

<http://www.skypoint.com/members/jlogajan>

• *Infinite Energy* home page:

<http://www.mit.edu:8001/people/rei/CFdir/CFhome.html>
<http://ourworld.compuserve.com/homepages/JedRothwell/>

COLD FUSION TIMES



MORE REPORTS:

News and Developments from California, Washington, D.C., Illinois, Massachusetts, New Mexico, Rhode Island, Texas, Utah, China, Denmark, England, France, India, Italy, Japan, Netherlands, Russia, Sri Lanka, United Kingdom. Plus copious material science and engineering news.

Big Happenings.

This issue of the **COLD FUSION TIMES** features reports from ICCF-6, the US Navy (two), Italy, China, and Japan. ICCF-6 meeting summaries include material data, helium production information, confirmations from France of the "boil-off experiment", and continued reports demonstrating low energy emissions, and helium production linked to the detection of anomalous excess heat. Here the French confirm Fleischmann and Pons.

REPRODUCTION OF FLEISCHMANN AND PONS EXPERIMENTS

G. Lonchamp, L. Bonnetain, P. Hicter, CEREM/CENG, ENSEEG Saint Martin d'Heres, France The French Atomic Energy Commission, in association with the ENSEEG (Ecole Nationale Supérieure d'Electrochimie et d'Electrometallurgie de Grenoble) started in

1993 a program of verification of the results of "cold fusion" published in 1993 by Fleischmann and Pons (1). Experiments have been performed in calorimeters identical to the ones used by Fleischmann and Pons (1). We point out here that these types of experiments can be analyzed in three temperature domains: - At low temperatures, below 70 C, excess enthalpy is the difference between the heat radiated to the water bath, and the enthalpy input due to electrolysis

- At intermediate temperatures, between 70 C and 99 C, excess enthalpy is the difference between the heat radiated towards the water bath plus the enthalpy contents of the gas stream plus the variation of enthalpy of the contents of the calorimeter because of the fast temperature variation observed) and the introduced electrical enthalpy

- In the boiling regime (without condensations, excess enthalpy is calculated from the difference between the total amount of water contained in the calorimeter evaporated and the theoretical quantity of water that should be evaporated by the energy introduced in the calorimeter (i.e. the enthalpy input due to electrolysis minus radiated enthalpy at boiling temperature).

Six calibration runs with platinum cathodes and 17 runs with different palladium type cathodes have been performed. At low temperature, 8 experiments have produced an excess energy rate between 1 and 5 %.

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COLD FUSION TIMES

P.O. Box 81135

Wellesley Hills, MA USA 02181

TO: